
Not All Spines Are Created Equal: How TotalSegmentator Fails on Lumbosacral Transitional Vertebrae

Anonymous Author(s)

Affiliation

Address

email

Abstract

Wrong-level spine surgery remains a persistent never-event, with the Lumbosacral Transitional Vertebra (LSTV, 5–35% prevalence) as its largest anatomic driver. We audit TotalSegmentator—the most widely deployed CT segmentation system and default back-end of 3D Slicer—and show that on lumbarization cases its output is anatomically indistinguishable from a wrong-level surgical plan: lacking an L6 class, it shifts lumbar labels caudally by one level, with junction-DSC dropping 29 points on Any-LSTV. No existing CT benchmark surfaces this—not TotalSegmentator’s own evaluation, not Li et al. [2025] (the per-class DSC frontier on the matched COLONOG split), and not VERIDAH (vertebra-labeling SOTA on a private CT cohort, operating on pre-localized crops with no pelvis). We contribute: (i) an LSTV-stratified evaluation protocol (per-class DSC, junction-DSC over a 40 mm L5/S1 window, per-class voxel confusion at the junction) translating the level-shift mechanism into quantifiable wrong-level surgical risk; (ii) CT-SPINOPELVIC1K—1,153 CT volumes across 802 patients with unified spinopelvic masks, 33 LSTV-positive cases across a 6-way phenotype taxonomy, and radiologist Castellvi typing (I–IV) on all 33; (iii) the first publicly-released merge-based LSTV-handling dual-spinopelvic CT segmenter, exceeding TotalSegmentator on sacrum (TotalSegmentator=0.817 zero-shot) and uniquely delivering both spine and pelvis in a single forward pass with released weights; (iv) an empirical isolation showing VERIDAH’s training-side L5/L6 merge eliminates the collision driving TotalSegmentator’s level shift but exposes a residual L4/last_lumbar collision on count-style sacralization, motivating VERIDAH’s sequence predictor as the necessary disambiguation component. CTSPINOPELVIC1K, the protocol, and 5-fold checkpoints are released publicly.

1 Introduction

Wrong-level spine surgery is a persistent never-event in operative spine care [Konin and Walz, 2010]. Despite intraoperative imaging and surgical-pause checklists, the most common cause—failure to correctly count vertebral levels in the presence of an LSTV—continues to account for the majority of reported wrong-level lumbar incidents [Carrino et al., 2011]. The mechanism is well documented: a surgeon counts lumbar bodies upward from the sacrum, finds five (because lumbarization presents a sixth lumbar body that has not been recognized), and labels the bottom body as L5 when it is anatomically L6. Surgery proceeds one level caudal to the intended target.

Modern radiology workflows increasingly route automated segmentation as an upstream substrate for clinical decision-making. The most widely deployed CT segmentation system in this role is TotalSegmentator [Wasserthal et al., 2023]: an nnU-Net trained on 1,204 routine CTs to delineate 104

anatomic structures, the default segmentation back-end of 3D Slicer, with thousands of clinical and research installations. The original paper reports an aggregate test-set Dice of 0.943 across all 104 classes and qualitatively documents two spine failure modes affecting neighboring-vertebrae label confusion (1% of cases) and neighboring-rib confusion (12%). TotalSegmentator is neither the only public CT spine segmenter nor the per-class DSC frontier on the matched CTSpine1K + CTPelvic1K (COLONOG) split (§2); it is, however, the model surgeons and downstream tools consume. This paper concerns one anatomic class on which TotalSegmentator does not perform at its headline level, and the consequences when it does not.

The clinical claim. On lumbarized spines, TotalSegmentator produces an output pattern that is, when consumed by surgical planning pipelines or by surgeons confirming operative levels, *indistinguishable from a wrong-level surgical plan*. Its label vocabulary contains L1–L5 but no L6 class. On a lumbarized spine (six lumbar bodies plus sacrum), the model distributes its five lumbar labels across the six bodies by shifting caudally by one anatomic level: the body that is anatomically L6 receives the L5 label, the body that is anatomically L5 receives the L4 label, and so on. A surgeon targeting the disc between predicted-L4 and predicted-L5 lands on the anatomical L5–L6 disc, one level caudal to the intended L4–L5 target. TotalSegmentator reproduces the canonical LSTV-induced miscount automatically and silently.

The architectural alternative: VERIDAH. Recent work [Möller et al., 2026] addresses transitional vertebra labeling via a two-stage architecture: a 3D DenseNet169 classifier on per-vertebra crops (four heads: label, region, transition, visibility), followed by a constrained sequence predictor that imposes anatomical-count constraints (4–6 lumbar bodies; the predicted sequence may end in two consecutive L5 labels, the second interpreted as L6). The classifier is trained with L5/L6 (and T12/T13) labels merged for class-imbalance reasons; disambiguation is deferred to the sequence predictor. The architectural commitment is empirically validated (97.22% CT subject-level agreement) but has never been tested against direct semantic segmentation, and it is unknown whether a sufficiently expressive voxel-wise network trained at full resolution with the same merge requires the sequence predictor at all. We supply the missing test.

The evaluation gap. Neither TotalSegmentator, VerSe [Sekuboyina et al., 2021], CTSpine1K [Deng et al., 2021], nor CTPelvic1K [Liu et al., 2021] report LSTV-stratified performance. Earlier CT labelers explicitly excluded the failure mode [Sekuboyina et al., 2020]. VERIDAH reports anomaly performance on a private cohort without isolating semantic-segmentation contribution. The result is a measurement vacuum: a production model emits a confident, locally well-formed, caudally-shifted prediction on a 5–35%-prevalence congenital variant with no error signal at evaluation time.

Closing the gap requires hardware. Fitting both lumbar spine and pelvis at $0.75 \times 0.75 \times 0.80$ mm spacing and $256 \times 320 \times 320$ patch consumes ~ 100 GB GPU memory under nnUNetResEncUNetPlans_100G, available on H200 (140 GB HBM3e) but not on the L40S/A100/H100 GPUs that bracket TotalSegmentator’s training configuration. Neither VERIDAH nor SPINEPS train at this combined spinopelvic resolution.

Contributions.

- (i) **Wrong-level surgical risk in TS, quantified.** An LSTV-stratified evaluation protocol (per-class DSC, junction-DSC over a 40 mm L5/S1 window, per-class voxel confusion at the junction) measures TS’s systematic one-level caudal shift on lumbarization (L4=0.388, L5=0.291; 95%+ of GT-L6 voxels labeled L5 on representative cases, §4) and structural absence of L6 prediction across all evaluation cases—translating the level-shift mechanism into quantifiable wrong-level surgical risk.
- (ii) **CTSPINOPELVIC1K: a stratified spinopelvic benchmark.** 1,153 abdominal CT records across 802 patients, unified spinopelvic masks, 33 LSTV-positive patients across a 6-way subtype taxonomy (lumb $n = 14$; sacral_count $n = 9$; full sacralization $n = 6$; semi-sacralization $n = 2$; ambiguous $n = 2$), and a registration-free fusion pipeline that resolves a previously undocumented Y-axis coordinate-convention mismatch between CTSpine1K and CTPelvic1K. Both source datasets share the COLONOG cohort that underlies VerSe and CTSpine1K, but neither encodes pelvic context with LSTV stratification. Radiologist Castellvi typing (I–IV) on all 33 LSTV cases is released as an additional annotation layer.

- (iii) **First publicly-released merge-based LSTV-handling dual-spinopelvic CT segmenter.** A 5-fold CV nnU-Net 3d_fullres baseline on H200 hardware adopting VERIDAH’s training-side merge (last_lumbar = L5+L6) at the voxel level, with pelvic-class supervision in a single forward pass. We exceed TS on sacrum and uniquely deliver both spine and pelvis with released weights (Table 5); §2 details the reproducibility tiers separating openly-released from paper-only baselines.
- (iv) **Empirical isolation: merge alone is insufficient.** The merge eliminates the L5/L6 collision driving TS’s level shift but exposes a residual L4/last_lumbar collision on count-style sacralization. This residual collision empirically isolates the component of VERIDAH’s success attributable to its sequence predictor—a direction VERIDAH’s authors could not ablate.

2 Related Work

Wrong-level spine surgery and LSTV. [Konin and Walz, 2010, Carrino et al., 2011, Sekharappa et al., 2014] document LSTV-induced miscounting (5–35% prevalence) without addressing automated segmentation as a miscount-propagation pathway. Kwak et al. [2025] detect LSTV on plain radiographs (binary; different modality).

Production CT spine and pelvic segmentation. TotalSegmentator [Wasserthal et al., 2023], an nnU-Net trained on 1,204 CTs and the default 3D Slicer back-end, has no L6 class and no LSTV stratification; we audit it as the production system downstream pipelines actually consume, not as a per-class DSC frontier. Li et al. [2025] hold the published per-class DSC frontier on a COLONOG split (731 scans; 440/146/145) but release neither weights nor code (paper-only) and do not stratify by LSTV. Liu et al. [2025] release FracSegNet (PENGWIN+CTPelvic1K), a pelvic-only cascaded 3D nnU-Net providing sacrum and L/R hipbone; their qualitative note of sacralization degradation (their §4.3) corroborates the LSTV gap our benchmark surfaces. Table 5 compares all baselines.

Vertebra-labeling SOTA: VERIDAH. Möller et al. [2026] address transitional-vertebra *labeling* (distinct from segmentation): a four-head 3D DenseNet169 classifier on per-vertebra crops plus a constrained sequence predictor enforcing 4–6 lumbar bodies (allowing terminal double-L5 to encode L6). On a private CT cohort of 1,536 volumes VERIDAH reports 99.18% subject-level perfect labeling versus prior CT SOTA [Meng et al., 2023] 77.81%. Three properties matter for us: VERIDAH (i) requires upstream centroids (VerSe-derived models or SPINEPS [Möller et al., 2024], itself MRI-only); (ii) excludes pelvis; (iii) validates on a private CT cohort. Their ablation (Table 5) shows the sequence predictor adds substantial value, but the comparison against direct semantic segmentation on a public LSTV-stratified split is missing—the gap our CTSPINOPELVIC1K-trained baseline (§7) fills.

Benchmarks and evaluation rigor. VerSe [Sekuboyina et al., 2021] (460 CT volumes) is the only public CT benchmark whose label vocabulary includes L6 (label 25) but it is unstratified by LSTV and lacks pelvic GT; earlier CT labelers explicitly excluded the failure mode [Sekuboyina et al., 2020]. Stratified safety-critical reporting [Maier-Hein et al., 2024] motivates §6.

3 Failure-Mode Analysis: TS on LSTV

We evaluate TotalSegmentator v2 (total task) zero-shot at full resolution on the full CT-SPINOPELVIC1K matched cohort. TotalSegmentator was not trained on any CTSPINOPELVIC1K case, so every case is valid; this provides 1,152 of CTSPINOPELVIC1K’s 1,153 records (one record dropped due to TS inference failure) aggregated to 801 patients via patient-level merging, including 33 LSTV-positive patients. Full-resolution mode isolates the failure as a property of label vocabulary and training distribution rather than inference resolution.

Table 1 reports performance by phenotype. On normal anatomy ($n = 768$) TotalSegmentator performs at its published level. On Any-LSTV ($n = 33$) every metric drops; lumbarization collapses caudally (L1=0.878 $\rightarrow$ L4=0.388 $\rightarrow$ L5=0.291) with junction-DSC at 0.357. Count-style sacralization ($n = 9$) collapses uniformly across all lumbar classes (L1=0.265 down to L5=0.000). L6 predictions are structurally zero across all 1,152 cases.

Table 1: TS zero-shot stratified by LSTV phenotype on CTSPINOPELVIC1K, full-resolution mode. Junction DSC over a 40 mm L5/S1 axial window. L6 count: cases with any L6 voxel emitted (structurally zero by label vocabulary). †: junction analysis not computed for separate-mode records.

Subgroup	<i>n</i>	L1	L2	L3	L4	L5	Sac	Hips	Jxn DSC
All cases	801	0.911	0.906	0.913	0.914	0.913	0.817	0.929	0.818
Normal	768	0.921	0.918	0.929	0.936	0.938	0.817	0.929	0.824
Any LSTV	33	0.681	0.615	0.523	0.367	0.314	0.822	0.929	0.534
Lumbarization	14	0.878	0.853	0.690	0.388	0.291	0.833	0.928	0.357
Sacralization (count-style)	9	0.265	0.100	0.055	0.001	0.000	0.814	0.926	0.697
Semi-sacralization	2	0.951	0.951	0.951	0.955	0.956	0.824	0.937	†
Sacralization (full)	6	0.718	0.676	0.673	0.724	0.793	0.826	0.932	0.636
Ambiguous	2	0.952	0.951	0.882	0.678	0.374	0.771	0.932	†
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The level-shift mechanism. The asymmetry between lumbarization (severe, caudally progressive) and count-style sacralization (uniform collapse) follows from a single fact: TotalSegmentator’s output head has no L6 class. For 6-vertebra anatomies the model compresses 6 anatomic units into 5 labels by shifting label assignment caudally; the most rostral body remains L1 (escape into thoracic vertebrae), but every successive body is misaligned, with the final body labeled L5 when it is anatomically L6. For 4-vertebra anatomies the model emits standard L1–L5, overshooting the GT lumbar count and producing a +1-level offset uniformly. The level-shift mechanism is the segmentation analog of the surgical miscount [Konin and Walz, 2010].

4 Case Study: Wrong-Level Surgical Risk in TS Output

Aggregate metrics establish the population pattern; the case study below establishes the mechanism at the level a surgeon would consume the output. Token 215 is a lumbarization case (six anatomical lumbar bodies, separate spine and pelvic GT records merged at patient level). TS’s output head emits only L1–L5, so on a 6-vertebra anatomy the bottom-most body (anatomical L6) cannot receive its correct label. Table 2 reports per-class DSC: L1 and L2 are well-placed (0.954, 0.927), L3 is partially correct (0.594), and **L4, L5, and L6 all collapse to dice zero**—a clean signature of the level-shift, with no GT-L4 voxel correctly labeled, no GT-L5 voxel correctly labeled, and no L6 voxel emitted at all. The pelvic classes (sacrum, hips) remain at near-Normal accuracy, confirming that the failure is local to the lumbosacral count and not a general degradation of TS on this patient.

Table 2: Token 215 (lumbarization, 6 lumbar bodies): per-class TS zero-shot DSC. The L4–L5–L6 row is the empirical signature of the level-shift failure—all three collapse to dice zero because TS compresses 6 anatomical bodies into 5 labels. Pelvic classes are unaffected. † TS’s output head has no L6 label.

Class	L1	L2	L3	L4	L5	L6	Sacrum	Hip _L	Hip _R
DSC	0.954	0.927	0.594	0.000	0.000	0.000 †	0.817	0.922	0.926

Surgical translation. A surgeon planning an L4–L5 microdiscectomy and using the segmentation to confirm operative level identifies the disc between predicted-L4 (anatomical L5) and predicted-L5 (anatomical L6). The targeted disc is L5–L6, one level caudal to the L4–L5 intent. The output is locally well-formed: five compact vertebral predictions in roughly correct positions. The error is detectable only by counting bodies (six lumbar, not five), which downstream pipelines do not perform.

5 CTSPINOPELVIC1K: Construction and Composition

CTSPINOPELVIC1K fuses CT scans from COLONOG [Smith et al., 2015] with annotations from CTSpine1K [Deng et al., 2021] (VerSe-derived per-vertebra masks) and CTPelvic1K [Liu et al., 2021] (sacrum and bilateral hips). The same COLONOG cohort underlies VerSe and CTSpine1K;

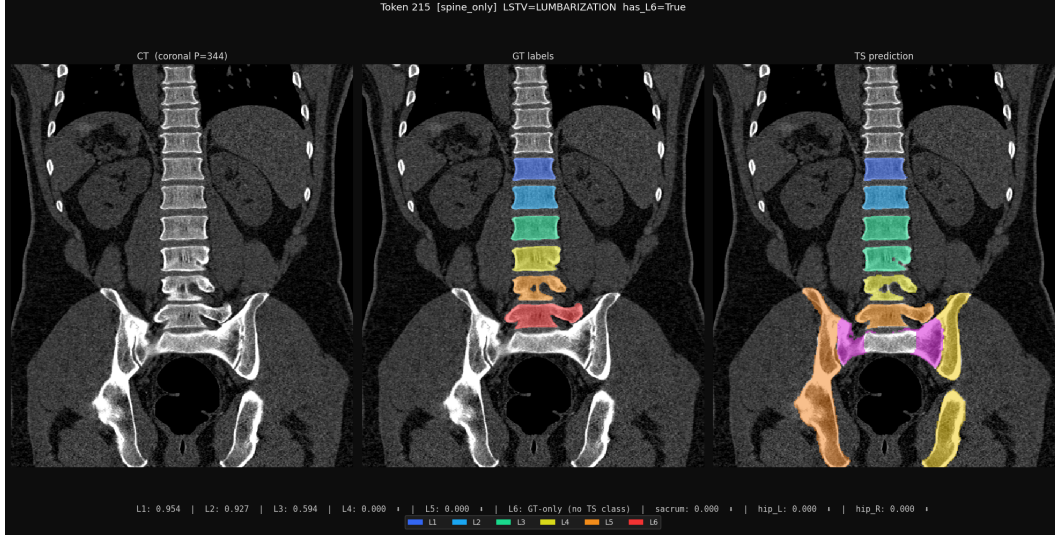


Figure 1: Token 215, lumbarization case (6 anatomical lumbar bodies). TS predictions overlaid on CT. The model compresses 6 bodies into 5 lumbar labels with a caudal cascade: the predicted-L3 region absorbs both anatomical L3 and L4 (yielding partial L3 dice 0.594, zero L4 dice); predicted-L4 sits over anatomical L5 (zero L5 dice); predicted-L5 sits over anatomical L6 (which has no label in TS’s output head; zero L6 dice). The output is locally well-formed—five compact vertebral predictions in approximately correct positions—but every label below L2 is one anatomical level off, with the bottom body unlabeled-as-L6 entirely. Quantified in Table 2.

Table 3: CTSPINOPELVIC1K composition; LSTV phenotype breakdown in Table 4.

Property	Value	LSTV cases
Patients matched (CTSPINOPELVIC1K)	802	—
Records matched (CTSPINOPELVIC1K)	1,153	33 patients (44 records)
Fused (spine + pelvis)	342	—
Separate (spine + pelvic)	352	—
Spine-only / Pelvic-only	90 / 20	—

CTSPINOPELVIC1K’s contribution is the patient-level fusion of CTSpine1K’s vertebra masks with CTPelvic1K’s pelvic masks plus the LSTV-stratification overlay neither source provides. Of 1,153 matched records, 342 carry fused (spine + pelvis) ground truth, 352 are separate-mode, 90 are spine-only, and 20 are pelvic-only (Table 3).

Patient-anchored fusion and coordinate correction. Patient identity is resolved via canonical UID matching (825/825 patients). Each mask is assigned to its source CT acquisition by intra-patient bone-HU maximization, $s^*(m) = \arg \max_{s \in \mathcal{S}_p} |\{v \in \mathcal{L}(m, s) : I_s(v) > 200 \text{ HU}\}| / |\mathcal{L}(m, s)|$, a metadata-free oracle since SeriesDescription fields are unreliable in TCIA releases (99.6% spine, 100% pelvic placement). CTSpine1K and CTPelvic1K were converted from DICOM with different dcm2niix invocations, producing a systematic Y-axis sign reflection in Nifti affines; we detect via affine-derived center sign and reverse along axis 1 with affine update. We adopt VERIDAH’s training-side merged-label scheme at the voxel level: 0 = bg, 1–4 = L1–L4, **5 = last_lumbar** (anatomical L5 in 5-vertebra spines; both L5 and L6 in 6-vertebra spines), 6 = sacrum, 7–8 = left/right hip, 9 = ignore.

Six-way LSTV taxonomy with dual-provenance labels. We retain both LSTV label sources separately and partition into six subtypes by combining filename-qualifier evidence (CTPelvic1K morphological criterion) with vertebral-counting evidence (CTSpine1K VerSe-label-presence heuristic): **normal**, **lumb** (6 vertebrae labeled or VerSe label 25 present), **sacr_count** (4 vertebrae labeled with sacralization flag), **semisacralization**, **sacralization** (both via filename qualifier), and **ambiguous** (cross-source disagreement). An earlier substring-match parser bug collapsed semisacralization into full sacralization—merging two anatomically distinct phenotypes whose failure modes diverge

sharply (full sacralization preserves the lumbar count and produces near-normal per-class DSC, whereas semisacralization introduces a partial transition that confuses the L5/S1 boundary). Any benchmark reporting an aggregate sacralization DSC under this collapse averages over two qualitatively different model behaviors; we identify and fix this. Without the corrected 6-way taxonomy, the qualitatively different failure patterns across LSTV variants (§3) are invisible.

Partial-annotation training protocol. Training only on fused records discards 70% of the data. We use nnU-Net v2’s `ignore_label` [Isensee et al., 2024] to train on all 1,153: voxels in unsupervised regions are written as label 9, contributing zero to Dice and cross-entropy. Spine classes (1–5) receive supervision from 784 records; pelvic classes (6–8) from 714. This is a $3.36\times$ effective expansion. Per-record subtype refinement gates training-time tags on supervised anatomy.

Castellvi annotation layer. The 6-way taxonomy captures the *count* dimension of LSTV. The Castellvi classification [Castellvi et al., 1984] captures the orthogonal *morphological* axis: Type I (dysplastic transverse process), II (pseudoarticulation), III (bony fusion), or IV (mixed II/III), each a (unilateral) or b (bilateral). It is the surgeon’s pre-operative wrong-level-risk indicator, and what a CT segmenter would need to predict to flag LSTV for review rather than silently emit a caudally-shifted label. Two resident radiologists jointly annotated all 33 LSTV cases by consensus (no independent inter-rater scoring), extending the schema—originally sacralization-specific—to score the bony junction uniformly across variants, with per-side a/b laterality.

Table 4: Castellvi distribution across all 33 LSTV cases of CTSPINOPELVIC1K, cross-tabulated by 6-way count taxonomy.

Subtype	Ia	Ib	IIa	IIb	IIIa	IIIb	IV	Total
Lumbarization	0	1	1	3	1	7	1	14
Sacralization (count-style)	0	0	0	0	0	9	0	9
Sacralization (full)	0	0	0	0	1	3	2	6
Semi-sacralization	0	0	1	0	1	0	0	2
Ambiguous	0	0	0	1	0	0	1	2
Total	0	1	2	4	3	19	4	33

Morphology-stratified failure interpretation. The cross-tabulation (Table 4) surfaces two findings with direct bearing on model behavior. (1) **All 9 count-style sacralization cases are Castellvi IIIb**—bilateral bony fusion—and 19/33 (58%) of LSTV overall is IIIb. The residual L4/last_lumbar collision (§8.1) therefore occurs precisely when the discriminating cue is locally observable: bony fusion of the bottom lumbar to sacrum is visually distinct from a normal L5–S1 junction, yet the segmenter still emits last_lumbar for those voxels. Direct semantic segmentation has not learned the Castellvi correlate of the count anomaly even when it is morphologically present in the input—an empirical argument for downstream Castellvi-aware reasoning at the lumbosacral junction (whether VERIDAH-style global count constraints or a dedicated Type III morphology head). (2) Only 1/33 cases is Type I (population prevalence 14–16% [Sekharappa et al., 2014]), reflecting source-dataset selection bias toward morphologically conspicuous presentations. Reported performance therefore characterizes the conspicuous regime; the clinically subtler Type I fraction is uncharacterized in this benchmark.

6 Evaluation Protocol

We define four metrics as the minimum required to attribute LSTV failure modes to wrong-level surgical risk [Maier-Hein et al., 2024]: (1) per-class DSC stratified by 6-way LSTV subgroup, exposing class-level degradation; (2) junction-DSC over a 40mm L5/S1 axial window, the clinical window for surgical planning; (3) per-class voxel confusion at the L5/S1 junction, which exposes structured failure modes (the level shift) that DSC alone cannot characterize; and (4) for the merged-label baseline, three focused per-epoch confusion blocks anchored on the L4/last_lumbar boundary that drives the residual collision (last_lumbar_GT_on_lumb, L4_GT_on_sacr_count, pred_last_lumbar_on_sacr_count) plus a specificity number (last_lumbar_specificity_on_sacr_count).

7 Spinopelvic Baseline

Architecture and training. nnU-Net v2 [Isensee et al., 2020, 2024] ResEnc-M planner, 3d_fullres, nnUNetResEncUNetPlans_100G (100 GB GPU budget). Spacing $0.80 \times 0.75 \times 0.75$ mm, patch $256 \times 320 \times 320$, batch 2; 7-stage residual encoder ($n_{\text{features}}=[32, 64, 128, 256, 320, 320, 320]$, $n_{\text{blocks}}=[1, 3, 4, 6, 6, 6, 6]$, 3^3 kernels, 2^3 strides, instance norm, deep supervision at 5 levels); 9-channel output (bg + 8 fg); ignore-label masked from loss. Class-reweighted DC_and_CE loss ($\mathbf{w}=[0.5, 1, 1, 1, 1, 1.5, 2.0, 1, 1]$; last_lumbar boost targets the clinically-important boundary, sacrum boost supports Castellvi precision); SGD Nesterov 0.99, $\lambda=3 \times 10^{-5}$, PolyLR 10^{-2} , 500 epochs (250 train + 50 val iters); 5-fold patient-stratified CV with 6-way LSTV stratification.

Two-tier LSTV emphasis. Queue-level oversampling: all five non-normal subtypes duplicated to 25% of per-fold training samples (matching VERIDAH’s $4 \times$ rate). **Class-reweighted CE** (above). No per-patch foreground biasing: under merged labels, biasing toward an L6 class is ill-defined, and biasing the L4/last_lumbar boundary on sac_count would confound the residual-collision measurement.

Hardware. H200 NVL (140 GB HBM3e), 24 CPU workers, preunpacking once per job to a 1.7 TB local-NVMe cache (replacing NFS reads that throttled at 21 MB/s under 5-fold contention). Per-epoch $\approx 400\text{--}700$ s; per-fold ≈ 3 days for 500 epochs. Reducing the GPU memory budget forces either smaller patch (loses spinopelvic context) or coarser spacing (loses Castellvi-defining detail); 100 GB is the smallest budget fitting both.

8 Results

Headline. The merged-label spinopelvic baseline closes the LSTV junction-DSC gap on lumbarization: junction-DSC rises from $\text{TS}=0.357$ to ours $=\mathbf{0.946} \pm 0.007$ (5-fold mean $\pm$ std; ensemble 0.950), a $\Delta=+\mathbf{58.9} \pm 0.7$ points (5-fold; ensemble +59.3 points). The model emits non-degenerate last_lumbar on all 3/3 held-out lumbarization test cases with per-case last_lumbar DSC of 0.698 ± 0.038 (5-fold; ensemble 0.678), eliminating the structural absence that drives TS’s level shift. A residual class collision persists on count-style sacralization: voxel-level confusion (§8.1, Block 2; $\sim 720\text{K}$ aggregated GT-L4 voxels across 5 folds) shows **82.7%** of GT-L4 voxels mis-predicted as last_lumbar, with per-case L4 DSC at 0.187 ± 0.337 (high cross-fold std driven by a single fold’s outlier; the voxel-level analysis (Table 7) is unaffected because its denominator is dominated by the other folds). §8.1 characterizes this as the empirical signature of direct semantic segmentation’s limit on transitional anatomy. On pelvic classes, our model exceeds TotalSegmentator on sacrum by $\Delta=+\mathbf{15.2} \pm 0.2$ points ($\text{TS}=0.817 \rightarrow \mathbf{0.969} \pm 0.002$ 5-fold; 0.970 ensemble) and establishes state-of-the-art for openly-released dual spinopelvic CT segmentation (Table 5).

LSTV-stratified per-subgroup view. Table 6 exposes the LSTV-induced degradation that aggregate per-class DSC masks; the residual L4 DSC drop on sac_count cases (§8.1) persists.

8.1 The Residual Class Collision

The merged scheme eliminates the L5/L6 collision driving TS’s wrong-level mechanism. Each fold’s checkpoint evaluates the same 174-case test split; voxel counts below pool 5 evaluations per case (15 case-fold pairs for lumb, 5 for sac_count). Across 5-fold predictions on lumb cases ($\sim 4.06\text{M}$ aggregated GT-last_lumbar voxels), **58.2%** of GT-last_lumbar voxels are correctly predicted as last_lumbar (Block 1), against TS’s structurally-zero last_lumbar emission. The residual 40.1% misallocation to L4 is the cost of label-merging at the joint, not a re-emergence of TS’s full collapse. The merge does, however, expose a residual collision at the L4/last_lumbar boundary on sac_count cases: GT-L4 voxels are predicted as last_lumbar in **82.7%** of voxels (Block 2; $\sim 720\text{K}$ GT-L4 voxels aggregated across 5 folds), and predicted-last_lumbar voxels on sac_count cases are anatomically L4 in **100.0%** (Block 3). last_lumbar specificity on sac_count = 0.778 ± 0.074 across folds.

Residual upper-lumbar caudal-shift cascade. While the merge restores correct labeling at the bottom-most boundary (Block 1), per-class DSC for upper lumbar on lumb cases shows a residual

Table 5: Per-class DSC ($\times 100$). **L5** is the unmerged anatomical L5 class; **last_lumbar** is the merged class L5+L6 (= anatomical L5 in 5-vertebra Normal anatomy, where the two columns measure the same voxel set under different label schemes). Hip_m merges LH+RH; we train per-side and report both. TotalSegmentator numbers are Normal-subgroup zero-shot on CTSPINOPELVIC1K (Table 1); its combined Hips is placed in Hip_m. Top-block numbers are reproduced from those works’ own test sets and are not directly comparable; bold marks best in column across all reported numbers. [†]Any-LSTV: case-weighted mean over the 7 LSTV test cases (per-subgroup numbers in Table 6).

Method	L1	L2	L3	L4	L5	last_lumbar	Sacrum	LH	RH	Hip _m
<i>Reproduced from prior work; not evaluated on CTSPINOPELVIC1K (Li et al. on COLONOG split, FracSegNet on PENGWIN test set)</i>										
3D U-Net	77.19	78.34	81.71	82.86	92.13	—	90.02	93.74	93.42	—
ResNet-50	80.29	76.41	80.62	83.77	91.05	—	88.37	94.72	94.01	—
ResNet-101	74.57	78.11	78.54	83.84	86.35	—	89.05	95.69	94.42	—
SwinUNETR	88.87	83.12	84.36	89.04	92.15	—	91.38	92.33	93.38	—
AttentionUNet	80.26	75.84	79.64	82.32	90.39	—	92.48	93.41	95.45	—
nnU-Net [Isensee et al., 2020]	89.35	91.37	92.41	93.37	95.74	—	96.72	95.50	96.78	—
Li et al. (CFFM+LT) [Li et al., 2025]	92.98	92.36	93.92	93.31	97.07	—	96.98	94.37	96.77	—
FracSegNet [Liu et al., 2025]	—	—	—	—	—	—	98.7	99.3	99.2	—
<i>Reproducible baseline (publicly-released weights), evaluated zero-shot on CTSPINOPELVIC1K</i>										
TotalSegmentator [Wasserthal et al., 2023]	92.1	91.8	92.9	93.6	93.8	—	81.7	—	—	92.9
<i>Ours: CTSPINOPELVIC1K, 5-fold ensemble on held-out test, partial-annotation supervision</i>										
Ours, Normal ($n=167$)	95.3	95.5	95.9	96.7	—	97.1	97.1	95.0	93.6	98.8
Ours, Any-LSTV ($n=7$) [†]	60.1	46.6	31.1	15.8	—	65.2	92.0	97.5	97.6	97.6

Table 6: CTSPINOPELVIC1K 5-fold ensemble on held-out test split ($n=174$), reported as cross-fold mean $\pm$ std. LL EMIT: cases emitting ≥ 100 last_lumbar voxels. Dashes: classes without GT in that subtype. Semi-sacralization absent from test split (§9). Δ vs. TS zero-shot (Table 1). * $n=1$ subgroup; $\pm$ omitted (cross-fold std on a single case reflects fold-checkpoint disagreement on one outlier rather than a population distribution; sacr_count L4 in particular shows high cross-fold variability with mean 0.187 and per-fold dice spanning near-zero to mid-range).

Subgroup	n	L1	L2	L3	L4	last_lumbar	Sac	Hip _m	Jxn DSC	LL emit
Normal	167	.953 $\pm$.008	.955 $\pm$.005	.959 $\pm$.004	.967 $\pm$.005	.971 $\pm$.005	.971 $\pm$.002	.988 $\pm$.001	.907 $\pm$.016	167/167
Lumbarization	3	.872 $\pm$.050	.611 $\pm$.123	.377 $\pm$.147	.213 $\pm$.114	.698$\pm$.038	.946 $\pm$.004	.960 $\pm$.002	.946$\pm$.007	3/3
Sacralization (count)*	1	.988	.959	.734	.187	—	—	—	—	1/1
Sacralization (full)	2	.001 $\pm$.002	.000 $\pm$.000	.001 $\pm$.002	.062 $\pm$.083	.583 $\pm$.136	.910 $\pm$.039	.989 $\pm$.000	.927 $\pm$.068	2/2
Ambiguous*	1	—	—	—	—	—	.862	1.000	—	1/1
Δ vs. TS (lumb)	—	−0.6	−24.2	−31.3	−17.5	N/A	+11.3	+3.2	+58.9	—
Δ vs. TS (sacr_count)	—	+72.3	+85.9	+67.9	+18.6	—	—	—	—	—

caudal-shift the merge does not resolve: L1=0.872, L2=0.611, L3=0.377, L4=0.213 (Table 6; std in table). The 5-vertebra positional prior anchors top-of-spine correctly but shifts labels caudally as the count discrepancy compounds. TS’s shift concentrates at the bottom (no L6 class); ours distributes it through the upper lumbar (the merge absorbs the bottom shift into last_lumbar)—on per-class L2–L4, TS is in fact better, trading upper-lumbar precision for bottom-anchor correctness. Junction-DSC, centered on L5/S1, closes the surgically critical gap (+58.9 pts) regardless: wrong-level risk is dominated by the bottom-most label. Fully resolving the upper cascade requires a sequence predictor with explicit count constraints—contribution (iv) realized empirically: merge alone is necessary but insufficient.

Interpretation. The model has learned the spinopelvic spatial relationship on normal anatomy and recovers the merged last_lumbar class on lumbarization. On count-style sacralization, however, the positional prior dominates: the bottom-most lumbar visible in a given patch is treated as last_lumbar regardless of the global vertebral count. The disambiguating signal (vertebra count from a known anchor) is non-local; a $256 \times 320 \times 320$ patch does not reliably contain both the caudal lumbosacral and rostral cervico-thoracic boundaries needed to count. This is a finding rather than a defect of the baseline—it executes the cleanest test of VERIDAH’s training-side merge in the semantic-segmentation setting and confirms the architectural hypothesis underlying its two-stage design: the merge addresses class imbalance, but distinguishing the last-lumbar voxels of a 4-vertebra spine from those of a 5-vertebra spine must be deferred to a downstream component with access to global anatomical structure.

Table 7: Residual class confusion under merged labels, focused on L4/last_lumbar. Voxel counts aggregated across 5 folds; percentages computed from the summed counts (denominator-weighted, not averaged percentages). Block 1: GT-last_lumbar on lumb (good signal, $\sim 4.06\text{M}$ voxels). Block 2: GT-L4 on sacrum_count (failure-mode signal, $\sim 720\text{K}$ voxels). Block 3: pred-last_lumbar on sacrum_count (inverse direction, $\sim 595\text{K}$ voxels).

Block	last_lumbar	sacrum	L4	background
Block 1: last_lumbar GT on lumb	58.2%	0.4%	40.1%	1.2%
Block 2: L4 GT on sacrum_count	82.7%	0.3%	14.9%	2.2%
Block 3: pred-last_lumbar on sacrum_count	—	0.0%	100.0%	0.0%

9 Discussion and Future Work

Generalization and hardware. Any spine segmentation model lacking an L6 class will exhibit the level-shift on lumbarization; the LSTV gap is the symptom, not a model-specific bug. Our 100 GB patch budget requires H200-class hardware; smaller budgets yield weaker gap closure and therefore residual wrong-level risk downstream.

Future work. (1) **VerSe expansion via sacrum pseudolabeling.** VerSe [Sekuboyina et al., 2021] contains 460 CT volumes labeling L6 (label 25) but no pelvis; our segmenter’s sacral DSC (0.969 ± 0.002 , 5-fold) supports sacrum pseudolabeling on VerSe for a 1,613-volume spinopelvic cohort. (2) **Two-stage integration with VERIDAH.** Our segmenter can replace VERIDAH’s VerSe-derived CT upstream, supplying sacrum and per-vertebra centroids in a single forward pass.

Limitations. COLONOG is a US screening cohort (ages 45–75); generalization to emergency/pediatric populations is not evaluated. LSTV labels are inherited from CTPelvic1K and CTSpine1K heuristics; Castellvi annotation covers all 33 LSTV cases but does not screen the normal cohort for Type I, and the source-dataset construction biases against morphologically-mild Type I presentations (§5). The held-out test split contains 7 of the 33 LSTV cases (3 lumb, 1 sacrum_count, 2 sacralization-morph, 1 ambiguous, 0 semi-sacralization); fold-level evaluation pools across the same 174-case test set, yielding 15 case-fold pairs for lumbarization and 5 for sacrum_count in voxel-confusion analysis but only the test-split n for case-level Dice (voxels within a case are highly correlated, so voxel counts in §8.1 represent dependent observations rather than independent samples). We do not run a same-compute unmerged 9-class baseline as a controlled ablation of the merge contribution; the merge’s role is inferred from comparison to TS (which has no L6 class) rather than a same-data ablation. TS is evaluated zero-shot, the correct comparison for using TS as a production system but undersells what it could achieve if retrained with an L6 class.

Implications. A production CT segmentation system that emits a confident, locally well-formed, caudally-shifted prediction on a 5–35%-prevalence congenital variant produces no error signal at inference time and, when consumed downstream, reproduces the canonical mechanism of wrong-level spine surgery. Any future spine segmentation model claiming readiness for surgical-planning integration should report per-subgroup DSC and per-class confusion at the L5/S1 junction on a stratified LSTV cohort.

Data and Code Availability

CTSPINOPELVIC1K is hosted on HuggingFace Datasets at <https://huggingface.co/datasets/anonymous-neurips-ED/CTSpinoPelvic1K> with both core and Responsible AI fields of the Croissant specification [Akhtar et al., 2024]; a 50-record sample covering all six LSTV subtypes is available at <https://huggingface.co/datasets/anonymous-neurips-ED/CTSpinoPelvic1K-Sample>. The evaluation protocol, data-construction pipeline, and trainer source are released under permissive license at <https://github.com/anonymous-neurips-ED/spinopelvic-seg>, accessible without credentialed access. Trained 5-fold checkpoints will be made available. All repositories are anonymized for double-blind review.

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output with ignore-label masking, the explicit class-reweighted DC+CE loss weight vector, SGD Nesterov ($\mu=0.99$, weight decay 3×10^{-5}), PolyLR (10^{-2}), 500 epochs (250 train + 50 val iterations), the two-tier LSTV emphasis (queue oversampling at 25% plus class reweighting), and 5-fold patient-stratified CV with 6-way LSTV stratification. §5 specifies the splits.

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Justification: Our segmenter is evaluated under 5-fold patient-stratified CV with 6-way LSTV stratification (§7); per-subgroup DSC are reported as 5-fold means $\pm$ standard deviation across folds in Table 6, and per-case DSC distributions are released alongside the per-fold checkpoints. Variability captured is across patient-stratified train/test splits. TS (Table 1) is evaluated zero-shot once per case with no per-fold variability (it was not retrained); we report per-case DSC means with n per subgroup so reviewers can recover dispersion from the released per-case CSV.

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8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: §7 (Hardware) specifies H200 NVL (140 GB HBM3e), 24 CPU workers, a 1.7 TB local-NVMe preunpack cache replacing throttled NFS reads, per-epoch $\approx 400\text{--}700$ s, and per-fold ≈ 3 days for 500 epochs (5-fold CV total ≈ 15 GPU-days). §1 documents that the 100 GB GPU-memory budget is the smallest configuration fitting both lumbar spine and pelvis at full spacing; smaller budgets force coarser patches or spacing and degrade LSTV gap closure.

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Answer: [Yes]

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: §9 (Implications) addresses both directions. Positive: surfacing a safety-critical, silent failure mode in the most widely deployed CT spine segmenter that produces an output indistinguishable from a wrong-level surgical plan on a 5–35%-prevalence congenital variant, and providing the LSTV-stratified protocol that any future model claiming surgical-planning readiness must report against. Negative: an LSTV-aware segmenter could itself be over-trusted by downstream pipelines if deployed without radiologist verification; we explicitly recommend that automated outputs require radiologist review for surgical planning, and the paper's central thesis is precisely that locally well-formed segmentation outputs cannot be assumed safe in the absence of LSTV-stratified evaluation.

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Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

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Justification: Neither the dataset nor the model is in the high-misuse-risk category targeted by this question. CTSPINOPELVIC1K is a strict derivative of fully-public TCIA cohorts (COLONOG, CTSpine1K, CTPelvic1K) and adds no new patient data, no PHI, and no scraped content. The released segmenter is a research-only spinopelvic model with no generative capability; its primary documented failure mode (residual L4/last_lumbar collision on count-style sacralization, §8.1) is itself one of the paper’s contributions, surfaced precisely to discourage uncritical clinical deployment.

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